INT387 : DATA ANALYTICS WITH POWER BI

PROJECT REPORT

(Project Semester January-May 2026)

Adult Income Census Analysis

Discipline of CSE
Lovely School of Computer Science & Engineering
Lovely Professional University, Phagwara

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CERTIFICATE

This is to certify that Venkata Nagaratna Viswesh Chakka bearing Registration No. 12402894 has successfully completed the course project titled **"Adult Income Census Data Analysis Using Power BI"** under my guidance and supervision for the course INT374. The work presented is original and meets the academic requirements.

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DECLARATION

I, Venkata Nagaratna Viswesh Chakka, student of B.Tech Computer Science & Engineering under CSE Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine. This project work is original and has not been submitted elsewhere for any academic award.

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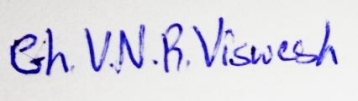

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1. INTRODUCTION

Income analytics plays a critical role in understanding socio-economic conditions, identifying income distribution patterns, and supporting data-driven decision-making in areas such as policy planning, workforce analysis, and economic research. With large and complex census datasets, traditional tabular reports often fail to provide meaningful insights. This project focuses on building an interactive **Income Analytics Dashboard** using Power BI, enabling efficient exploration of demographic trends, employment patterns, and income distribution.

The Income Analytics Dashboard has been developed using Microsoft Power BI with the objective of transforming complex census data into meaningful, interactive, and decision-oriented visual insights. Census datasets are inherently large, multi-dimensional, and difficult to interpret when presented in raw tabular form. Analysts, researchers, and decision-makers require intuitive tools that allow them to understand how factors such as age, education, occupation, and working hours influence income levels.

This project focuses on designing a comprehensive Power BI dashboard that provides a holistic view of income-related insights across multiple pages such as Overview, Education Analysis, and Age Analysis. The dashboard enables users to analyze income distribution, demographic characteristics, work patterns, and financial indicators such as capital gain through interactive visuals and filters. The emphasis of the project is not only on visualization but also on building a strong analytical foundation through proper data cleaning, data modeling, and DAX-based calculations.

The dashboard has been designed keeping non-technical users in mind. Therefore, clarity, readability, and logical flow of information were prioritized. Each visual element is intended to answer a specific analytical question, making the dashboard suitable for both exploratory analysis and high-level reporting.

2. OBJECTIVES OF THE PROJECT

The key objectives of this project are:

- To analyze Adult Census Income data using an interactive Power BI dashboard in order to identify income distribution patterns and socio-economic trends.
- To study demographic factors such as age, gender, education, and marital status to understand their impact on income levels.
- To analyze employment-related attributes including occupation, workclass, and hours-per-week to examine their relationship with income categories (>50K and ≤50K).
- To evaluate financial indicators such as capital gain and capital loss to understand their influence on higher income groups.
- To identify patterns and relationships between age groups and income distribution to determine which age segments have higher earning potential.
- To examine how working hours and employment type contribute to variations in income levels.
- To design an interactive and user-friendly Power BI dashboard with slicers, filters, and visualizations for dynamic exploration of income data.
- To transform raw census data into meaningful insights that support data-driven understanding of income trends and socio-economic behavior.

3. DATASET OVERVIEW

The dataset used in this project was collected from Kaggle, a widely used platform for data science and machine learning datasets. Kaggle provides access to diverse, community-driven, and publicly available datasets that are commonly used for data analysis, visualization, and predictive modeling projects. The dataset selected for this project contains detailed Adult Census Income data that enables comprehensive analysis of demographic, employment, and income-related attributes.

This dataset serves as the foundation for all analytical and visualization tasks performed in this project. It includes information related to individuals' demographic characteristics, employment details, and income categories, which helps in understanding socio-economic patterns and income distribution.

Link: https://www.kaggle.com/datasets/uciml/adult-census-income

Each row in the dataset represents an individual and contains multiple attributes describing different aspects of their profile. These attributes include age, workclass, education level, marital status, occupation, relationship status, gender, hours-per-week, capital gain, capital loss, and income category (>50K or ≤50K). These features are used to perform detailed analysis and create interactive visualizations in the Power BI dashboard.

4. FEATURE DESCRIPTION

The dataset used in this project consists of both categorical and numerical attributes that provide detailed information about individuals' demographic and socio-economic profiles. Each record represents an individual and includes features such as age, workclass, education, marital status, occupation, relationship, gender, native country, hours-per-week, capital gain, and capital loss.

These attributes are used to analyze patterns and relationships influencing income levels. Numerical features such as age, working hours, and capital gain help in identifying trends and measuring variations, while categorical features such as education, occupation, and workclass support grouping and comparison of different segments.

The dataset also includes a target variable, income, which classifies individuals into two categories: those earning more than 50K per year and those earning 50K or less. This variable serves as the foundation for all analysis and visualization in the dashboard.

Overall, these features are utilized to create interactive visualizations, apply filters and slicers, and transform raw census data into meaningful insights that support a better understanding of income distribution and socio-economic behavior.

5. TOOLS AND TECHNOLOGIES USED

- **Power BI Desktop:** Used as the primary tool for this project to create interactive dashboards and visual reports. It helped in presenting complex Spotify data in a clear and visually meaningful manner using charts, KPI cards, slicers, donut charts, scatter plots, and line charts. Power BI also enabled multi-page navigation, cross-filtering, and dynamic slicer interactions.

- **Power Query:** Used for data cleaning and transformation before analysis. It helped in handling missing values, removing duplicate records, standardizing column names, converting data types, and deriving new calculated columns such as duration in minutes and chill score. This step ensured that the dataset was accurate, consistent, and ready for analysis.

- **DAX (Data Analysis Expressions):** Used to create calculated measures and KPIs required for the dashboard. It helped in computing metrics such as average popularity by genre, total listening hours, high-popularity track counts, explicit content percentages, and composite audio feature scores. DAX made the dashboard dynamic and responsive to filters and slicers.

6. DATA PREPARATION AND CLEANING

Data preparation and cleaning were carried out using Power Query in Power BI and formed one of the most critical phases of this project. The raw Adult Census dataset contained missing values, inconsistencies, and formatting issues that could affect analytical accuracy if not handled properly.

Duplicate records were checked and removed to avoid incorrect aggregation. Missing values in important columns such as workclass, occupation, and native country were carefully examined. Where necessary, missing or unknown values were handled appropriately to ensure data integrity while retaining useful information for analysis.

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6.1 Understanding the Dataset Structure

The initial step involved understanding the overall structure of the dataset. This included reviewing the number of records, columns, data types, and the role of each variable. Columns were classified into categorical variables (such as workclass, education, occupation, and gender) and numerical variables (such as age, hours-per-week, capital gain, and capital loss).

This step helped in identifying relevant features for income analysis and removing unnecessary or redundant columns to simplify the data model.

6.2 Handling Missing and Inconsistent Values

The dataset contained missing and inconsistent values, particularly in columns such as workclass, occupation, and native country. Some entries were marked as unknown or had incomplete information.

To address this issue, missing values were analyzed based on their importance. Records with critical missing information were either handled by assigning appropriate categories (such as “Unknown”) or excluded where necessary. This ensured that the analysis remained accurate and reliable.

6.3 Standardization of Categorical Fields

Categorical attributes such as education, occupation, and workclass contained variations in naming and formatting. Inconsistencies in capitalization or spelling could lead to incorrect grouping during analysis.

To resolve this, categorical fields were standardized by applying consistent naming conventions and formatting. This step ensured accurate grouping, filtering, and comparison across different categories in the dashboard.

6.4 Removal of Duplicate Records

The dataset was checked for duplicate entries that could distort analysis results. Duplicate records can lead to incorrect counts and misinterpretation of trends.

Using Power Query, duplicate rows were identified and removed to ensure that each individual record was unique. This helped maintain the accuracy of aggregated values and visualizations

6.5 Data Type Validation and Formatting

Another important step was validating and correcting data types. Some columns were not in the appropriate format, which could cause errors during calculations.

Numerical columns such as age, hours-per-week, capital gain, and capital loss were ensured to be in proper numeric format. Categorical columns were formatted as text. This step ensured smooth execution of calculations, filters, and visualizations within the dashboard.

6.6 Creation of Derived Metrics

After cleaning the dataset, derived metrics were created to enhance analysis and provide deeper insights.

Important derived fields included:
• **Age Group:** Categorizing individuals into groups such as 20–29, 30–39, etc., for better analysis
• **Income Flag:** A simplified classification of income categories (>50K and ≤50K)
• **Work Category Insights:** Grouping employment types for easier comparison
• **Average Hours Metrics:** Used to analyze workload patterns across different groups

These derived columns were created using DAX and Power BI transformations to support meaningful visualizations.

6.7 Outlier Detection and Validation

The dataset was analyzed to identify unusual values in numerical fields such as age, hours-per-week, and capital gain.

Outliers were carefully reviewed rather than removed blindly. For example, higher capital gain values indicated high-income individuals, which are valid and important for analysis. Similarly, variations in working hours reflected real-world employment patterns.

This step ensured that meaningful data was retained while maintaining data quality.

6.8 Final Data Validation for Dashboard Readiness

Before building the dashboard, a final validation was performed to ensure data consistency and accuracy. Aggregated values were verified, and sample records were cross-checked to confirm correctness.

This final step ensured that the dataset was clean, structured, and ready for analysis, enabling the creation of reliable and interactive Power BI visualizations.

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7. ANALYSIS ON DATASET

7.1 Overview and KPI Analysis

Introduction

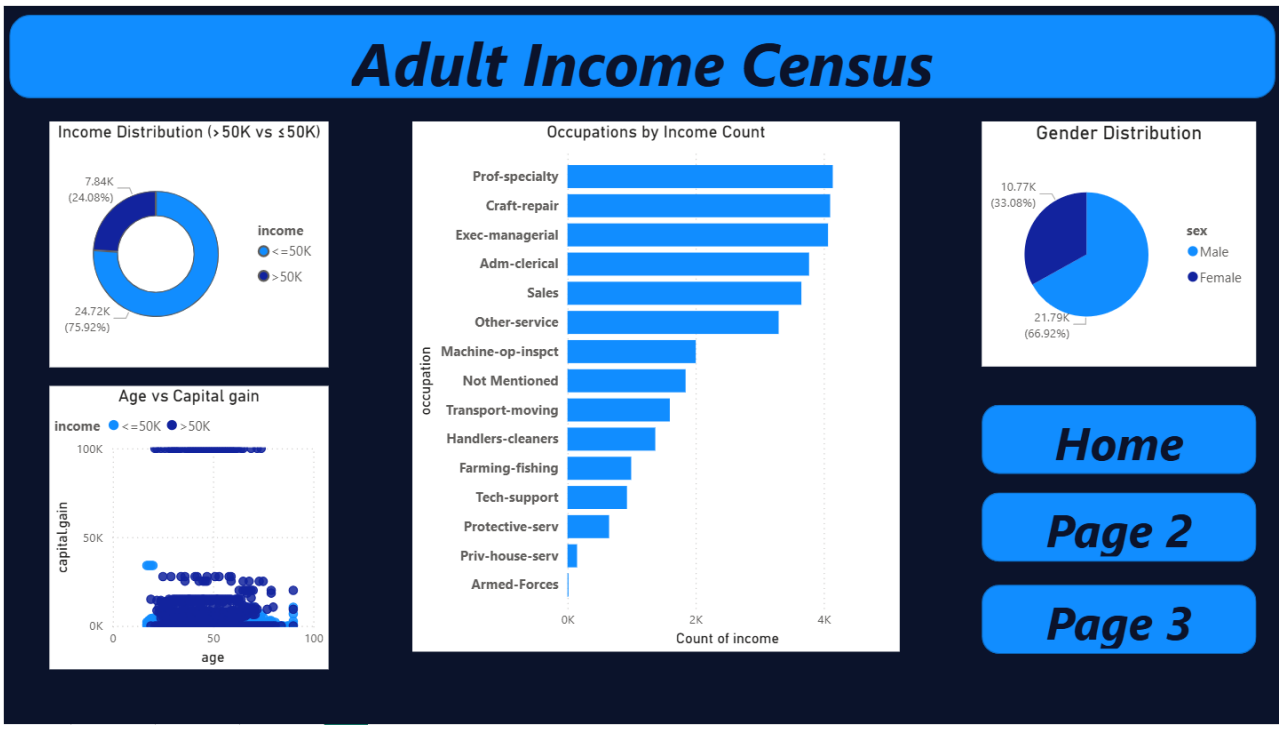
The Overview page serves as the central summary of the entire Adult Census Income dataset. It presents high-level KPI cards that provide a quick understanding of the dataset and income distribution patterns. These key metrics — total individuals, count of high-income and low-income groups, and average working hours — form the foundation for deeper analysis.

Metrics Used.

- **Total Individuals:** Total number of records in the dataset
- **High Income Count (>50K):** Number of individuals earning more than 50K
- **Low Income Count (≤50K):** Number of individuals earning 50K or less
- **Average Hours-per-week:** Average working hours across all individuals
- **Income Distribution by Education:** Comparison of income categories across education levels
- **Income Distribution by Occupation:** Income comparison across different job roles
- **Income Distribution by Workclass:** Income comparison across employment types

Analysis Results

The overview analysis shows that the majority of individuals fall under the ≤50K income category, indicating a skewed income distribution. Higher education levels such as Bachelors and Masters are strongly associated with higher income groups. Occupations in professional and managerial roles show a greater proportion of individuals earning above 50K, while labor-intensive and service roles tend to fall in the lower income category.



7.2 Demographic Analysis

Introduction

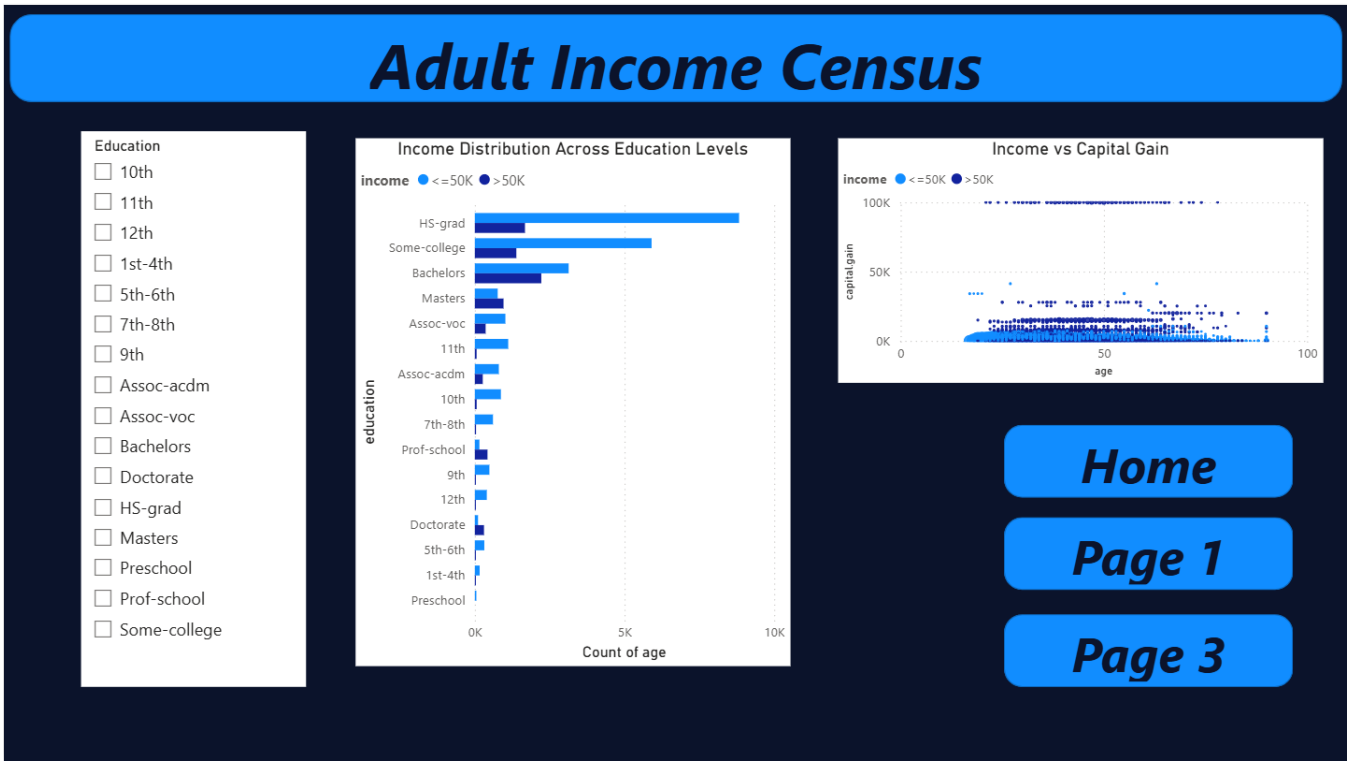
Demographic analysis focuses on understanding how personal attributes such as age, gender, and marital status influence income levels. This helps in identifying which demographic groups have higher earning potential and how income varies across population segments.

Metrics Used

- **Income Distribution by Age Group:** Comparison of income across different age ranges
- **Income Distribution by Gender:** Male vs Female income comparison
- **Average Age by Income Group:** Average age of high and low-income individuals

Analysis Results

The demographic analysis reveals that individuals in the age group of 30–50 have the highest proportion of high-income earners. Income levels tend to increase with age up to a certain point and then stabilize. Gender-based analysis shows that male individuals have a higher representation in the >50K category. Married individuals, particularly those in stable family roles, show higher income levels compared to unmarried individuals.



7.3 Employment and Work Analysis

Introduction

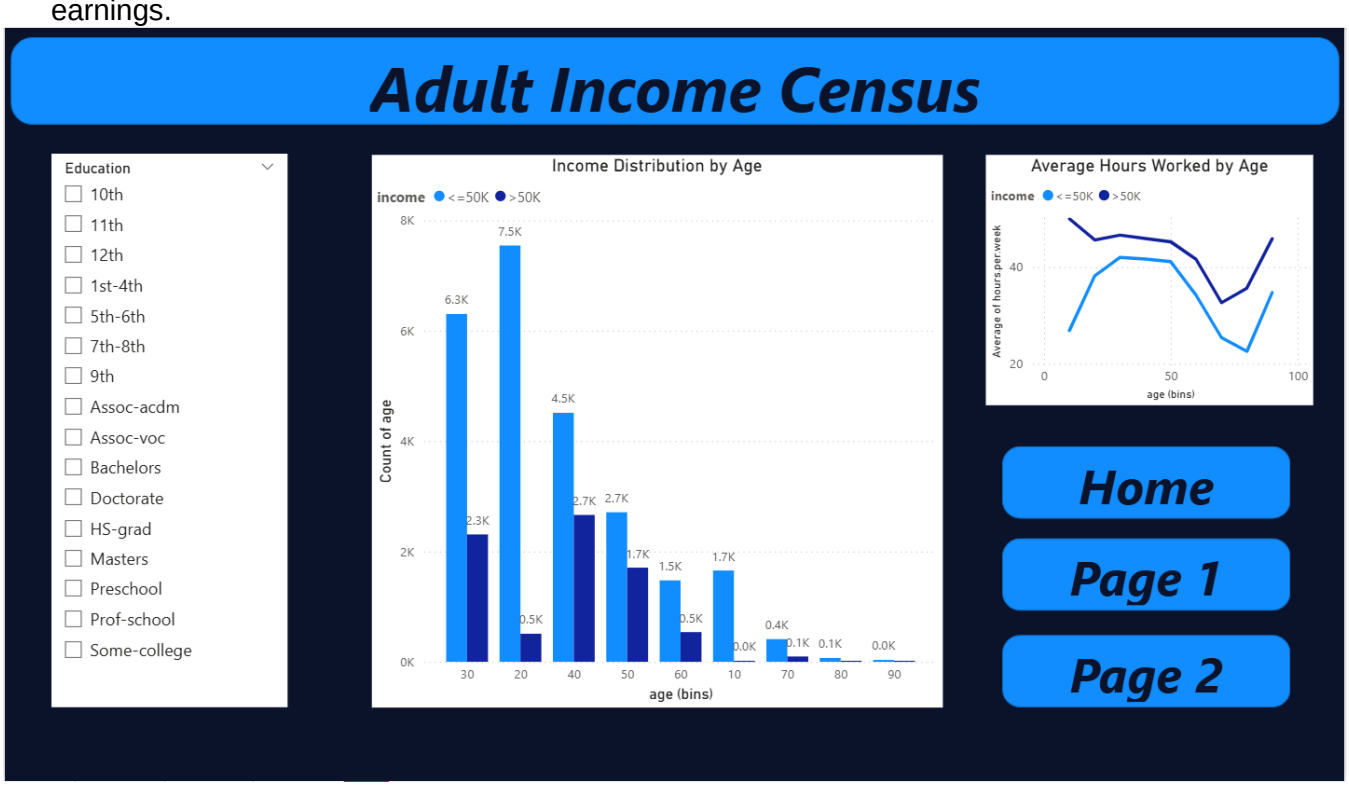
Employment analysis examines how job-related factors such as occupation, workclass, and working hours influence income levels. This helps in identifying the types of jobs and work conditions associated with higher earnings.

Metrics Used

- **Income Distribution by Occupation:** Income comparison across job roles
- **Income Distribution by Workclass:** Private, government, and self-employed comparison
- **Average Hours-per-week by Income:** Comparison of working hours between income groups
- **Work Hours vs Income Scatter Plot:** Relationship between working hours and income

Analysis Results

The analysis indicates that individuals working in executive, managerial, and technical occupations are more likely to earn above 50K. Self-employed individuals and those in private sector jobs also show higher income potential. Additionally, individuals working longer hours per week tend to fall into the higher income category, suggesting a correlation between workload and earnings.



8. DASHBOARD DESIGN AND INTERACTIVITY

The Power BI dashboard developed in this project was designed with a strong emphasis on clarity, usability, and professional presentation. The dashboard adopts a clean and minimal theme with balanced use of colors to ensure readability and reduce visual clutter. The design focuses on presenting socio-economic insights in an intuitive and easy-to-understand manner, making it suitable for both technical and non-technical users.
The dashboard is structured across multiple pages such as Overview, Demographic Analysis, Employment Analysis, and Financial Insights. Each page is designed to answer specific analytical questions while maintaining consistency in layout, color scheme, and visual elements.

8.1 Use of Interactive Slicers and Filters

To enhance interactivity, slicers were added for important dimensions such as Education, Occupation, Workclass, Gender, and Income category. These slicers allow users to dynamically filter the dataset and analyze patterns for specific groups.

As slicer selections are changed, all visuals update automatically. This dynamic behavior allows users to explore different scenarios, such as analyzing income distribution for a specific education level or comparing income patterns across different occupations, without reloading the dashboard.

8.2 Drill-Down and Hierarchical Navigation

The dashboard supports drill-down functionality, enabling users to move from summarized views to more detailed analysis. For example, users can start with income distribution by education and drill down further into specific occupation or workclass categories. This feature allows deeper analysis without overcrowding the dashboard with too many visuals and helps maintain a clean and structured layout.

8.3 Cross-Filtering and Cross-Highlighting

Cross-filtering and cross-highlighting features were enabled across all visuals. Selecting a category in one chart (such as a specific occupation or education level) automatically updates and highlights related data in other visuals. This interconnected behavior enhances the exploratory experience and helps users understand relationships between demographic, employment, and financial factors influencing income.

8.4 Visual Design and Theme Selection

A simple and professional theme was used for the dashboard, focusing on clarity and readability. A combination of neutral backgrounds with contrasting colors was applied to highlight key metrics and categories.

Different colors were used consistently to represent income groups (>50K and ≤50K), making comparisons easy. Fonts, spacing, alignment, and chart formatting were kept uniform across all pages to maintain a clean and structured appearance.

8.5 Selection of Visuals

KPI cards provide a quick summary of total individuals and income distribution. Bar and column charts are used for comparing income across education, occupation, and workclass. Line charts help analyze trends across age groups, while scatter plots show relationships between variables such as age, working hours, and capital gain..

8.6 Multi-Page Navigation

The dashboard includes structured navigation across multiple pages to improve user experience. Each page focuses on a specific aspect of analysis, allowing users to move easily between different sections. This organized layout ensures that users can explore data step-by-step without confusion, making the dashboard intuitive, user-friendly, and suitable for effective data storytelling.

9. KEY INSIGHTS AND FINDINGS

The analysis of Adult Census Income data using the Power BI dashboard resulted in several important insights related to demographic patterns, employment characteristics, and financial indicators influencing income levels.

- The dataset contains over 233K tracks across 27 genres and 15K unique artists, with an average popularity score of 41, indicating a wide spread of mainstream and niche content.
- The dataset contains a large number of individual records with a clear imbalance between income groups, where the majority of individuals fall under the ≤50K category, indicating an uneven income distribution.
- Education plays a significant role in income levels. Individuals with higher education qualifications such as Bachelors, Masters, and Doctorate are more likely to earn above 50K, while lower education levels are associated with lower income categories.
- Age-wise analysis shows that individuals in the 30–50 age group have the highest concentration of high-income earners. Income tends to increase with age up to mid-career stages and stabilizes afterward.
- Occupation and workclass have a strong influence on income. Professional, managerial, and technical roles show a higher proportion of individuals earning above 50K, while service and labor-intensive jobs are concentrated in the lower income category.
- Working hours are positively correlated with income levels. Individuals working longer hours per week are more likely to fall into the higher income group, indicating the importance of workload in determining earnings.
- Financial indicators such as capital gain strongly differentiate income groups. Individuals with higher capital gains are significantly more likely to belong to the >50K category, highlighting the role of investment income in overall earnings.
- Gender-based analysis reveals that male individuals have a higher representation in the high-income category, indicating a disparity in income distribution across genders.
- Derived features such as age groups and income categories helped simplify analysis and provided clearer insights into socio-economic trends, making the dashboard more effective for decision-making.

10. CONCLUSION

The Income Analytics Dashboard developed using Power BI successfully transforms raw Adult Census data into meaningful and interactive insights. The project demonstrates how data visualization and analytical techniques can be used to understand complex socio-economic patterns and income distribution.

Through this dashboard, key factors influencing income such as education, age, occupation, workclass, working hours, and financial indicators were analyzed effectively. The results highlight that higher education levels, professional occupations, increased working hours, and higher capital gains are strongly associated with individuals earning above 50K. Additionally, demographic factors such as age and gender also play a significant role in income variation..

The use of interactive visuals, slicers, and multi-page design enables users to explore the data dynamically and gain insights from different perspectives. Overall, the project successfully meets its objectives by providing a clear, user-friendly, and insight-driven analytical solution for income analysis.

11. FUTURE SCOPE

While the current dashboard provides valuable insights, there are several opportunities to enhance and extend this project further:

- **Predictive Analysis:** Machine learning models can be integrated to predict income levels based on demographic and employment features.
- **Real-Time Data Integration:** Connecting the dashboard to live or updated datasets can provide more relevant and real-time insights.
- **Advanced Visualizations:** Additional advanced visuals such as decomposition trees and AI-based insights in Power BI can be implemented for deeper analysis.
- **Geographical Analysis:** Incorporating location-based data can help analyze regional income disparities and trends.
- **User Personalization:** Role-based dashboards can be created to provide customized views for different types of users such as researchers, policymakers, or business analysts.
- **Data Expansion:** Including more features such as industry type, work experience, or economic indicators can improve the depth and accuracy of analysis.

12. REFERENCES

- Kaggle — Adult Census Income Dataset. Open dataset containing demographic, employment, and income-related attributes used for analysis. Available at: https://www.kaggle.com/datasets/uciml/adult-census-income
- • UCI Machine Learning Repository — Adult Data Set. Original source of the census income dataset widely used for data analysis and machine learning research. Available at: https://archive.ics.uci.edu/ml/datasets/adult
- • Microsoft Power BI Documentation — Official documentation used for understanding Power BI features, data modeling, DAX functions, and dashboard development practices. Available at: https://learn.microsoft.com/en-us/power-bi/
- • DAX Guide — Comprehensive reference for Data Analysis Expressions (DAX) functions used in Power BI for creating calculated columns and measures. Available at: https://dax.guide/

13. LINKEDIN

LinkedIn post sharing the Spotify Audio Insights Power BI Dashboard project.

Link: https://www.linkedin.com/posts/viswesh-ch1210_powerbi-dataanalytics-datavisualization-ugcPost-7453430576700817408-63Vs?utm_source=social_share_send&utm_medium=android_app&rcm=ACoAAFHLWXcBEjMwwkUmsVNchY2bUQfNi0TJbqc&utm_campaign=copy_link

14. DRIVE LINK

To view the Power BI Dashboard File and Dataset:

Link: https://drive.google.com/drive/folders/1wyMZPqgeK90j2-

lPR7A6GTOlhNwlkrPe?usp=sharing